5.3.9.3 Low-speed internal (LSI) RC oscillator

Table 38. LSI oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSI}	LSI frequency	V_{DD} = 3.3 V, T_J = 25°C	31.4[1]	32	32.6[1]	kHz
		T_J = -40 to 130°C	29.4[2]		33.6[2]	
		T_J = -40 to 140°C	28.6[2]	-	33.6[2]	
$t_{su(LSI)}$[3]	LSI oscillator startup time	-	-	80	130	µs
$t_{stab(LSI)}$[3]	LSI oscillator stabilization time (5% of final value)	-	-	120	170	
$I_{DD(LSI)}$[3]	LSI oscillator power consumption	-	-	130	280	nA

1. *Guaranteed by test production.*
2. *Evaluated by characterization - Not tested in production.*
3. *Specified by design - Not tested in production.*

5.3.10 Flash memory characteristics

Table 39. Flash memory characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
I_{DD}	Supply current	Word program	-	1	-	mA
		Page erase	-	0.8	-	
		Mass erase	-	0.8	-	

Table 40. Flash memory programming

Symbol	Parameter	Conditions	Min(1)	Typ	Max[1]	Unit
tprog	Word programming time	128 bits (user area)	-	20.0	160.0	µs
		16 bits (OTP / EDATA area)	-	20.0	160.0	
t_{ERASE} 8KB	Page (8 KB) erase time	-	-	2.0	2.1	ms
t_{ERASE} 2KB	Page (2 KB) erase time	-	-	2.0	2.1	
t_{ME}	Bank Mass erase time	-	-	96.0		
	Mass erase time	-	-	192.0	200.0	
V_{prog}	Programming voltage	-	2.65	-	3.6	V

1. *Evaluated by characterization - Not tested in production.*

Table 41. Flash memory user and EDATA endurance and data retention

Symbol	Parameter	Conditions	Min[1]	Unit
N_{END}	Endurance	T_J = -40 to 140 °C	10	Kcycle
t_{RET}	Data retention	1 Kcycle at T_J = 125 °C	10	Year
		1 Kcycle at T_J = 85 °C	30	
		10 Kcycle at T_J = 55 °C	30	

1. *Evaluated by characterization - Not tested in production.*